



Addressing high-performance data sparsity in metasurface inverse design using multi-objective optimization and diffusion probabilistic models

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Abstract: Recent advancements in deep learning, particularly generative networks capable of producing high-freedom structures, have significantly enhanced the precise generation of meta-atoms. However, these methodologies typically rely on an abundance of high-performance data, which remains scarce in many practical design scenarios. To bridge this gap, our study introduces what we believe to be a novel approach that synergistically combines multi-objective optimization algorithms with an enhanced diffusion model featuring an attention mechanism, termed MetaDiffusion-Att. Using the complex design task of dual-polarized, wide-angle incidence, and broadband low-emissivity electromagnetic glass as an application example, we demonstrate the effectiveness of our method through qualitative and quantitative experiments. The introduced multi-objective optimization method significantly captures more high-performance samples while ensuring high degrees of freedom, compared to currently widely used generic dataset construction methods. The MetaDiffusion-Att model, improved by the introduced attention mechanism, significantly outperforms conventional WGAN-GP and conditional VAE methods in generation accuracy and quality under small datasets. Furthermore, the proposed method exhibits extrapolation capabilities, generating new structures with performance surpassing that of the dataset, further enriching the design space. This framework provides a promising solution for the inverse design of metasurfaces in challenging scenarios with sparse high-performance samples.

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1. Introduction

With the growing global focus on low-carbon emissions, low-emissivity (Low-E) glass has become widely adopted for its energy-saving benefits in architectural facades and automotive glass. This type of glass significantly reduces infrared heat radiation through its multiple metal coatings; however, it poses challenges for millimeter-wave signal transmission [1], which is crucial for beyond 5G/6G communication technologies. Recent studies have begun to explore the integration of electromagnetic metasurface with Low-E glass, designing electromagnetic windows that manipulate wave propagation via artificially designed surface geometries to meet both low-carbon requirements and electromagnetic transmission [2].

Current design methodologies for these electromagnetic windows are typically limited to trial-and-error iterations guided by the designer's intuition, involving the selection of conventional unit structures and parameter scanning to achieve specific functionalities [3–5]. These methods are constrained by the degrees of freedom in the structures and the experience of the designers,

often resulting in inefficient designs for complex tasks with significant non-linearities [6]. Optimization-based inverse design methods, such as adjoint-based topological optimization [7], genetic algorithms [8], and ant-colony optimization [9], although improving precision for specific device designs, are prone to local optima and incur high computational costs due to individual optimizations for different targeted unit cells.

In contrast, deep learning has proven effective in exploring global design spaces [10], enabling the rapid generation of meta-atom structures that meet S-parameters requirements and can be reused after a single training [11–15]. Recent studies have further explored the integration of physics principles or domain knowledge with deep learning approaches to enhance design efficiency and accuracy [16–18]. In particular, generative networks, capable of producing high-degree-of-freedom structures, have found widespread application in metasurface design tasks. Among them, methods based on variational autoencoders (VAE) [19] suffer from poor reconstruction and blurry generation results due to the need to balance reconstruction error and Kullback-Leibler (KL) divergence error during the training process [6,20]. On the other hand, generative adversarial network (GAN)-based [21] methods have inherent instability due to the adversarial process requiring the generator and discriminator networks to have comparable strengths and reach a dynamic Nash equilibrium, resulting in difficulties in model design and training, which in turn affects the model's generation accuracy and quality [22,23]. Recently, diffusion-based methods [24], which learn the noise diffusion process and effectively denoise to achieve targeted results, have been shown to significantly outperform GAN-based and VAE-based methods in terms of generation quality and accuracy across various domains, including text-to-image generation [25,26], image super-resolution [27], and precise inverse design of metasurfaces [28].

The effectiveness of above methods often depends on abundant, high-quality samples for the target task, which are essential for constructing balanced training datasets [29]. However, in complex tasks such as designing dual-polarized, wide-angle incidence, and broadband low-emissivity electromagnetic glass, the design space is characterized by a scarcity of high-performance samples. This scarcity challenges conventional high-degree-of-freedom dataset collection methods like needle-drop method [30–32], which struggle to provide sufficient high-performance examples, resulting in severely imbalanced dataset distributions. Such imbalance significantly hinders the ability of neural networks to effectively learn and generate high-performance devices. While some studies have employed generative networks for dataset augmentation [33–35], these approaches are constrained by limited extrapolation capabilities. This constraint allows only for gradual expansion of distribution coverage, making them effective primarily for mildly imbalanced datasets but inadequate for severely imbalanced scenarios [36]. Consequently, the efficient application of generative networks to design tasks with rare or severely imbalanced high-performance samples remains underexplored and warrants further investigation [37].

In this work, faced with the challenge of sparse high-performance samples in complex design tasks, we propose a novel design approach that combines multi-objective optimization algorithms and an enhanced diffusion model, as shown in Fig. 1. Initially, the multi-objective optimization algorithm effectively captures sparse high-performance samples, significantly mitigating dataset imbalance and enhancing overall quality. Subsequently, we incorporate an attention mechanism [38] into the diffusion model, forming the improved MetaDiffusion-Att network. This network not only enhances feature learning capabilities but also improves the precision and stability of the model in generating complex samples. Particularly, our proposed method excels in scenarios where design requirements are stringent and high-performance samples are difficult to obtain. When addressing the design task of Low-E glass, even under small dataset with extremely sparse high-performance samples, our method not only demonstrates superior accuracy and generation capabilities compared to conventional WGAN-GP [39,40] and cVAE [41] but also

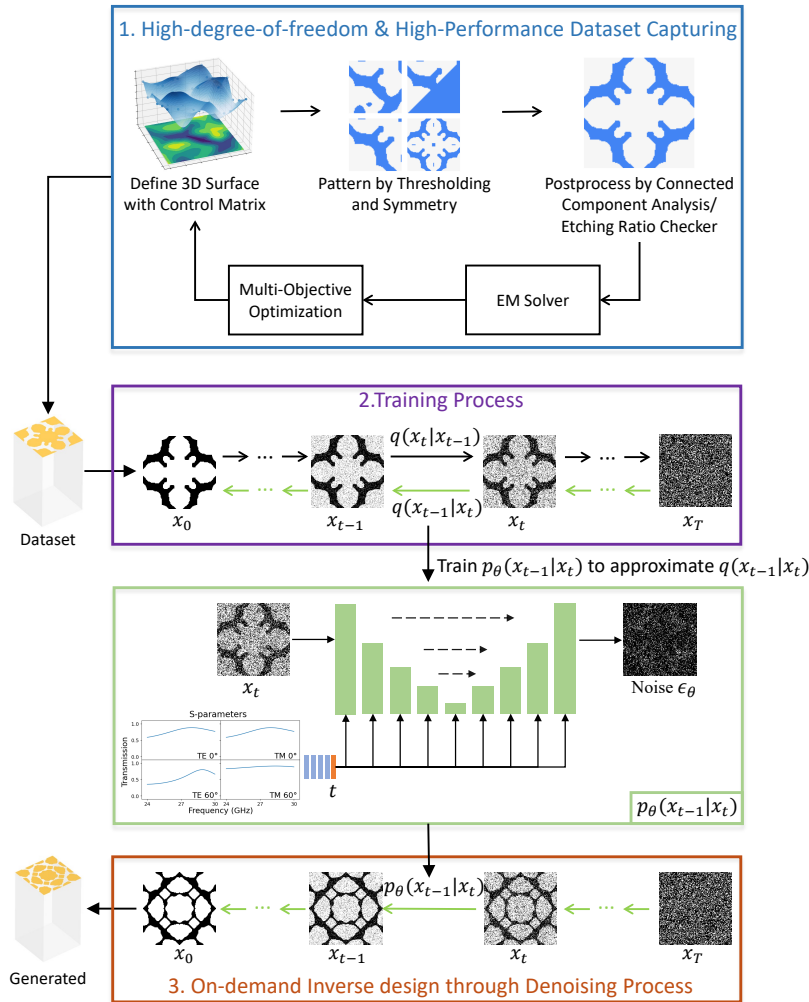


Fig. 1. Overview of proposed inverse design pipeline. (a) Blue Panel: Structure generation and multi-objective optimization process. A 6×6 control matrix is used as optimization variable to define a 3D surface, which is then processed to obtain the structure pattern. An EM solver simulates the transmission response for the multi-objective optimization algorithm to improve control matrix for the next optimization round. (b) Purple Panel: Schematic diagram of the training process. From left to right, the process of gradually adding noise to the original meta-atom is shown (black arrow), while from right to left, the process of denoising by gradually removing noise from samples on a Gaussian distribution to regenerate meta-atoms is shown (green arrow). (c) Green Panel: A neural network is used to approximate the posterior probability $q(x_{t-1}|x_t)$ required in the denoising process. By inputting the noisy meta-atom x_t , the noise ϵ_θ to be removed is predicted. The transmission amplitude and time step t are used as conditions and are integrated into the neural network after feature extraction to control conditional denoising. (d) Orange Panel: On-demand inverse design process. The well-trained neural networks are used in denoising process to generate new meta-atoms that meet the requirements, by starting with random noise and performing directional denoising according to the S-parameters conditions.

effectively generates designs that meet the requirements of dual polarization and large angle of incidence on demand. Through detailed experimental studies, we validate the effectiveness of the method in complex design spaces and its strong adaptability in high-performance sample-sparse environments.

2. Methods

2.1. Problem setup

To represent complex design scenarios with sparse high-performance samples, we consider an industry-relevant design requirement consisting of a 5mm thick glass substrate with a dielectric constant of 5.05, coated with a transparent Low-E film. This film is composed of approximately 20 microscopic layers, including three layers of silver, which notably enhance thermal infrared isolation—a critical advantage over designs with fewer silver layers. However, the tri-silver Low-E film's low sheet resistance (1 Ohm/sq) hinders millimeter-wave signal transmission.

In order to ensure the silver layers maintain their infrared thermal insulation capabilities while enhancing signal transmission, the objective is set to achieve on-demand design of dual-polarized transmission response for any desired bandwidth within the 24-30 GHz range and incident angles spanning 0-60 degrees, by etching no more than 30% of the film layer. The 30% etching ratio limitation is critical, as excessive etching would compromise the silver coating's thermal infrared isolation properties, which are a key performance attribute of Low-E glass. However, the presence of multiple silver layers, necessary for effective thermal insulation, inherently limits the achievable transmission performance in this frequency range. This leads to a scarcity of high-performance samples in the design space, posing a significant challenge for conventional data-driven inverse design methods. The design is represented by a 128×128 -pixel binary matrix, where 0 indicates etched regions and 1 represents the preserved tri-silver film layers. The structure is symmetrical along the x-y axes and diagonal, with a unit period set to 2.8 mm.

2.2. Multi-objective optimization for capturing sparse high-performance samples

To address the severely distribution imbalance caused by sparse high-performance samples while maintaining a universally applicable inverse design pipeline, we forgo relying on expert experience to introduce conventional structures. Instead, we leverage a novel approach that combines a versatile method for generating high-degree-of-freedom structures with the multi-objective optimization algorithm NSGA-II, as shown in Fig. 2. This synergistic approach effectively captures the required high-performance samples, crucial for addressing the data scarcity challenge and mitigating the initial severe imbalance. Notably, this approach ensures our design pipeline's broad applicability, proving suitable not only for the current Low-E electromagnetic glass design but also for a wide variety of design tasks facing sparse high-performance samples. This adaptability is particularly significant in the field of metamaterial design, offering a more universal solution to the persistent challenges of data scarcity and imbalance.

As illustrated in Fig. 2 and the blue panel of Fig. 1, we employ a structure generation method that utilizes a 6×6 control matrix to ultimately produce a high-degree-of-freedom pattern matrix. This method begins by generating a three-dimensional surface from the control matrix [42]. A cross-section is then extracted by applying a height threshold along the z-axis, retaining pixels with heights exceeding 0.5. This cross-section is then mirrored to introduce symmetry to the design. To ensure manufacturability, post-processing algorithms, such as connected region analysis, are employed to remove any isolated pixels. Finally, the etching ratio of the resulting pattern is evaluated against the design requirements. If the criteria are not met, the process iterates until a satisfactory pattern matrix is achieved. This approach, starting from a compact 6×6 control matrix, allows for the exploration of a vast design space with a high degree of

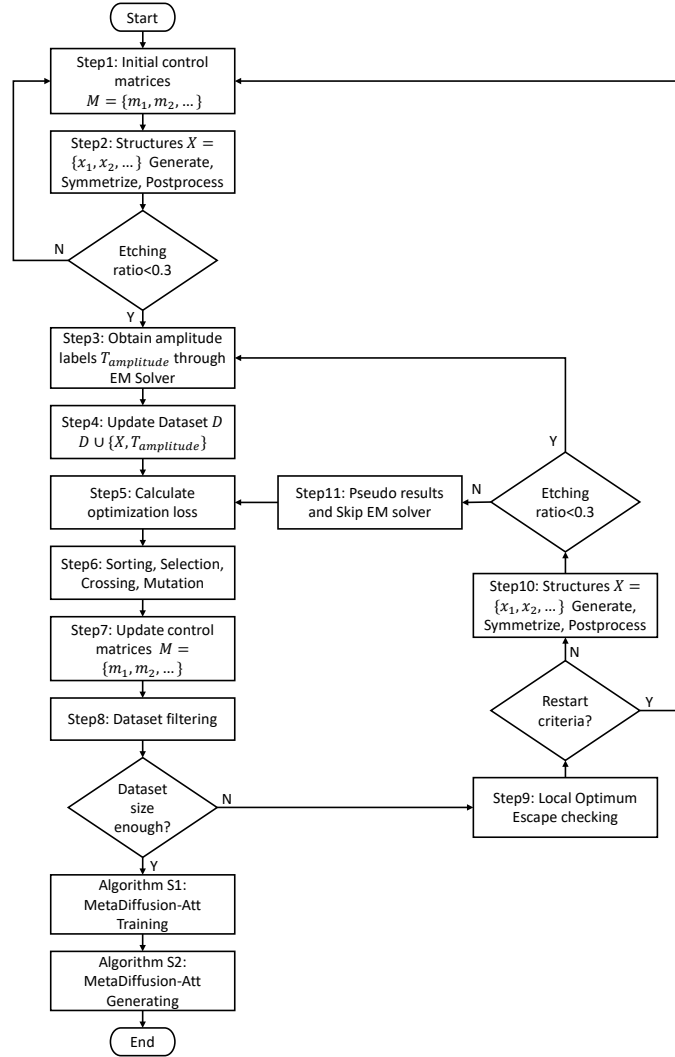


Fig. 2. Flowchart of the proposed approach. The detailed explanation of each step is illustrated in [Supplement 1 S1](#).

freedom while presenting a manageable number of optimization variables to the algorithm, thereby streamlining the optimization process.

During optimization, the 6×6 control matrix serves as the variable, and the pattern matrix is input into CST for simulation. This simulation measures the dual-polarized transmission response under large-angle incidences, serving as the optimization evaluation metric. The optimization targets are -2 dB bandwidths at 0 degrees and 60 degrees, with an etching ratio of 0.3 as a constraint:

$$\min f_1 = -\text{Bandwidth}(\text{Band}_{TE0^\circ} \cap \text{Band}_{TM0^\circ}) \quad (1)$$

$$\min f_2 = -\text{Bandwidth}(\text{Band}_{TE60^\circ} \cap \text{Band}_{TM60^\circ}) \quad (2)$$

$$\text{s.t. } g_1(\text{pattern}) = \text{ratio}_{\text{etching}} - 0.3 \leq 0 \quad (3)$$

During the multi-objective optimization process, the algorithm performs multiple rounds of optimization on the control matrix through fast non-dominated sorting, selection, crossover, and

mutation processes. To prevent the optimization algorithm from converging to local optima and generating similar structures, we implement structural and S-parameter similarity as restart indicators. This strategy, coupled with data deduplication filtering, ensures a diverse and high-quality dataset.

It's important to note that our approach does not aim to exhaustively search the entire feasible solution space to find the global optima through multi-objective optimization alone. Instead, we utilize multi-objective optimization primarily to construct a diverse dataset of relatively high-performance samples. This strategy leverages our neural network's ability to efficiently learn and generalize patterns from this diverse set of samples. Consequently, we avoid the need for numerous time-consuming initializations and iterations typically required to find global optima through optimization alone, significantly reducing the number of electromagnetic simulations required. By employing strategies like non-dominated sorting, we identify samples that exhibit strong performance under various conditions, providing a rich foundation for our neural network to learn from.

2.3. Proposed MetaDiffusion-Att model

While our pipeline combining multi-objective optimization algorithms with free-form structure generation strategy effectively captures high-performance samples and mitigates distribution imbalance, the inherent sparsity of high-performance samples in the design space still results in a relatively small dataset. To address this small sample size challenge, we incorporate an attention mechanism into the diffusion model, resulting in an enhanced MetaDiffusion-Att network. This improved model not only augments feature learning capabilities but also enhances the precision and stability in on-demand generating of high-degree-of-freedom Low-E electromagnetic glass patterns. By doing so, we effectively leverage the limited high-performance samples, enabling our model to extract more nuanced information and generate more accurate results despite the constraints of a smaller dataset.

As shown in Fig. 1, the model operates through two main processes: a forward diffusion process and a reverse denoising process. The forward diffusion process systematically incorporates noise into the original Low-E structural matrix x_0 through a Markov chain:

$$q(x_{1:T}|x_0) := \prod_{t=1}^T q(x_t|x_{t-1}) \quad (4)$$

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad (5)$$

Here, the variance β_t ranges from 0 to 0.02 and increases linearly from the initial timestep $\beta_1 = 1e - 4$ to the final timestep $\beta_T = 0.02$, indicating the amount of noise added at each step.

During the reverse denoising process, the MetaDiffusion-Att network is trained to characterize the inverse of the forward diffusion Markov chain $q(x_{t-1}|x_t)$. Hence, the problem transforms into minimizing the KL divergence between the true distribution $q(x_{t-1}|x_t)$ and the distribution $p_\theta(x_{t-1}|x_t)$ characterized by the neural network. This minimization process is achieved by using the network $\epsilon_\theta(x_t, t)$ to predict the noise ϵ_t added during the forward process:

$$\min_{\theta} L(\theta) = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2] \quad (6)$$

As shown in Fig. 3, we employ a generic encoder-decoder U-Net architecture to predict noise ϵ_t for a given noisy structural matrix x_t . This architecture comprises a series of encoders with residual blocks and convolutional layers for down-sampling, followed by interpolation layers and residual blocks for up-sampling. Skip connections between the encoder and decoder mitigate information loss during down-sampling, ensuring effective feature extraction.

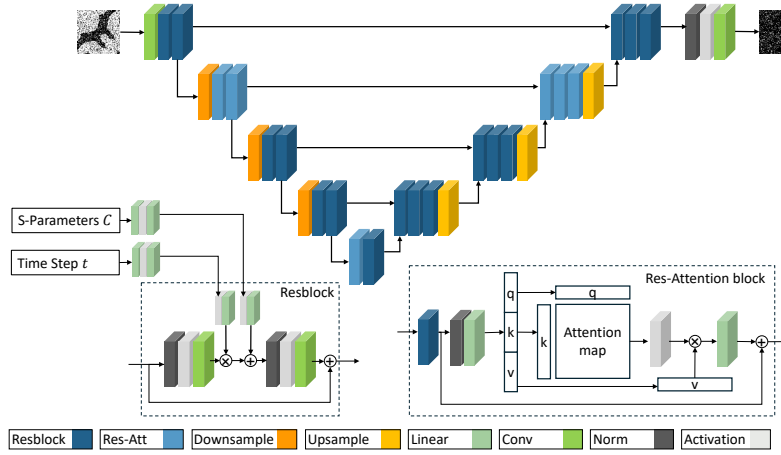


Fig. 3. MetaDiffusion-Attention network architecture.

To enhance the generated structures' accuracy in meeting transmission amplitude objectives, we implement a classifier-free training strategy [43]. This strategy involves randomly selecting 10% of the data samples and masking their conditions with a specific label "0". This implicit guiding encourages the model to focus on the remaining condition labels, leading to more accurate structure generation.

During the on-demand generation phase, we combine the conditional estimate $\epsilon_{\theta}(x_t, t, c)$ and the unconditional estimate $\epsilon_{\theta}(x_t, t, 0)$ to form the network's noise prediction. This combination, controlled by the ratio w , allows us to balance generation accuracy with diversity. Finally, new Low-E structures that meet the transmission response requirements can be generated through a gradual denoising process:

$$\tilde{\epsilon}_{\theta}(x_t, t, c) = (1 + w)\epsilon_{\theta}(x_t, t, c) - w\epsilon_{\theta}(x_t, t, 0) \quad (7)$$

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \tilde{\epsilon}_{\theta}(x_t, t, c) \right) + \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \cdot z \quad (8)$$

where $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, $\alpha_t = 1 - \beta_t$ and $z \sim \mathcal{N}(0, 1)$.

2.4. Amplitude information fusion and attention mechanism

Given the complexity and challenges inherent in achieving on-demand design of Low-E electromagnetic glass with large-angle and wide-bandwidth transmission frequency response targets, we introduce two key improvements to our previous work MetaDiffusion model. The first enhancement involves a more comprehensive integration of input features throughout the network. Specifically, we uniformly sample the transmission amplitude at 51 frequency points between 24-30 GHz for both 0° and 60° incident angles and both TE and TM modes, resulting in a 204-dimensional feature vector c . Three fully connected layers extract features from this 204-dimensional vector c and the time step t . In contrast to the original MetaDiffusion model, where features were only inputted into the decoder, MetaDiffusion-Att integrates these features into the residual blocks. This approach ensures effective fusion of the transmission amplitude conditions throughout the network.

Another key improvement in our MetaDiffusion-Att model is the incorporation of an attention mechanism, depicted as the Res-Attention block in the network diagram. This mechanism allows the model to selectively focus on the most relevant parts of the input features when predicting the

noise at each denoising step. Specifically, the output X from the residual blocks is reshaped and passed through Linear layers, producing three tensors: query $q = X \cdot w^q$, key $k = X \cdot w^k$, and value $v = X \cdot w^v$. The attention mechanism can then be represented by the following formula:

$$\text{Attention}(q, k, v) = \text{softmax} \left(\frac{qk^T}{\sqrt{d_k}} \right) \cdot v \quad (9)$$

The process of multiplying q and k to obtain the attention map can be understood as self-correlation of the input X to the self-attention mechanism, thereby learning which information in the features should be assigned more weight. The scaling factor $\sqrt{d_k}$ in the attention calculation ensures stable training by preventing vanishing gradients. Finally, the attention block calculates a weighted sum of the input features. This can be thought of as the network "paying attention" to certain aspects of the input while ignoring less important information. This selective attention mechanism enables the model to learn more complex relationships between the structure and its electromagnetic response, leading to improved generation accuracy.

3. Results and discussion

In this section, we present a comprehensive evaluation of our proposed methodology, focusing on three key aspects: (1) the effectiveness of our multi-objective-optimization-based high-degree-of-freedom dataset construction pipeline; (2) the generation accuracy and quality of the proposed MetaDiffusion-Att network; and (3) the performance of our integrated method—combining both the dataset construction pipeline and neural network—in the complex task of inverse designing dual-polarized, wide-angle incidence, and broadband low-emissivity electromagnetic glass. Our experiments are implemented in Python, utilizing Pymoo for optimization, CST Microwave Studio for electromagnetic simulations on an Intel i9-13900k setup, and PyTorch 2.2 on an Nvidia A6000 GPU for neural network operations.

We begin by demonstrating the effectiveness of our proposed dataset construction pipeline in addressing the challenge of high-performance sample scarcity inherent to this design task. Figure 4 illustrates the distribution of collected samples based on their -2 dB bandwidth at 0 and 60-degree incident angles. The blue dots represent 5211 samples collected using the widely adopted needle-drop method, a random high-degree-of-freedom structure generation approach. As evident from the distributions at the top and right of Fig. 4, the majority of samples exhibit limited bandwidth performance, concentrating around 0 GHz or demonstrating bandwidth only at normal incidence (0 degrees). Obtaining devices with significant bandwidth at both normal and large-angle incidences proves exceptionally challenging, highlighting the extreme sparsity of high-performance samples within the design space. Consequently, such conventional random generation methods struggle to construct balanced datasets in scenarios characterized by high-performance sample scarcity. This imbalance can negatively impact neural network training, hindering its ability to effectively learn the underlying design principles and generate high-performance devices.

In stark contrast, our proposed dataset construction pipeline demonstrates a superior ability to capture high-performance samples, as visualized by the orange lines in Fig. 4. This targeted sampling strategy effectively mitigates the severe imbalance inherent in the design space, resulting in a significantly more uniform data distribution. This improved dataset forms the foundation for subsequent evaluations of our MetaDiffusion-Att network's accuracy and effectiveness in generating high-performance designs.

Building upon this improved sampling approach, we retained samples from the optimization process and performed de-duplication based on structural similarity, ultimately collecting a total of 15,016 samples for our dataset. This dataset was then randomly divided into training, validation, and test sets in an 8:1:1 ratio. It's worth noting that given the complexity of the design space, this is considered a relatively small dataset. Each meta-atom structure in the dataset is

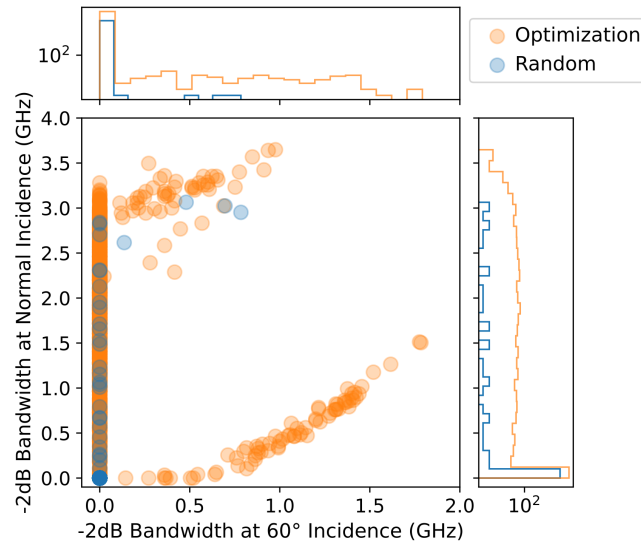


Fig. 4. Comparison of datasets obtained via conventional random sampling (needle-drop method) and our proposed multi-objective optimization pipeline. The main subplot (bottom-left) visualizes samples based on their bandwidths at normal and 60-degree incidence angles. Accompanying histograms (top and right) illustrate the data distribution. Blue points represent samples from the needle-drop method; orange points indicate samples from multi-objective optimization.

represented by a 64×64 Boolean matrix in the upper left corner, which incorporates symmetry information. To characterize the electromagnetic properties, we uniformly sampled 204 points from the corresponding transmission responses for 0 degree and 60 degree incident angles with TE and TM polarizations between 24-30 GHz, using these as condition labels.

During the training and testing comparison of the generative neural network, it is necessary to generate targets on the validation and test sets, followed by forward simulation of the generated results to obtain the corresponding transmission responses, thereby quantitatively evaluating the network's accuracy performance. However, using simulation methods such as FDTD is highly time-consuming, with speeds far slower than the generation speed of the network, making it challenging to handle the evaluation of large amounts of data during the neural network training and testing process. Therefore, we utilize PNN as a rapid solver [44], enabling accurate spectral response predictions of meta-atoms within milliseconds. We have validated through experiments that PNN serves as a reliable surrogate solver, with an L1 loss of 0.0066 on the test set. (See the [Supplement 1](#) for more experimental results.)

To thoroughly demonstrate the accuracy and stability of our proposed method's on-demand direct generation capabilities, we benchmark its performance against our previous work, MetaDiffusion, as well as two representative direct generation methods: WGAN-GP and conditional VAE (cVAE). WGAN-GP enhances the stability of traditional GANs by incorporating the Wasserstein distance and introducing a gradient penalty to ensure Lipschitz continuity, thereby providing more stable and reliable training. The cVAE, a VAE-based method, utilizes S-parameters as inputs to the decoder, enabling the direct generation of meta-atoms.

After binarizing the generation outcomes of the four models (using a threshold of 0.5), we employ the surrogate solver for forward calculations to determine the corresponding transmission

amplitudes. We use the mean absolute error (MAE) as the metric for accuracy:

$$L = \frac{1}{n} \sum_{i=1}^n |T_{\text{label}}(i) - T_{\text{gen}}(i)| \quad (10)$$

where $T_{\text{label}}(i)$ is the label transmission amplitude at the i -th sample point and $T_{\text{gen}}(i)$ is the transmission amplitude corresponding to the generated result at the i -th sample point. The above calculation includes the 204 sample points for the 0° TE/ 0° TM/ 60° TE/ 60° TM incidences.

During the training phase, we train each model for 500 epochs, and all four models gradually converge within the 500 epochs. We select the best-performing epochs from the validation set as the final well-trained model for final generation. Figure 5 and Fig. 6 visually present randomly selected generation results from the four models. Each method's row displays the direct model output and the structure after binarization. Finally, the binarized structure is forward simulated to compare the transmission amplitude of the generated structure with the target. From the figure, we can see that all four methods can generate structures that roughly conform to the target transmission trend according to the required transmission conditions. However, our proposed MetaDiffusion-Att method has the highest accuracy among the four methods. The original MetaDiffusion method also performs well, generating clear structures, but with slightly less accuracy compared to MetaDiffusion-Att. Nonetheless, it still outperforms WGAN-GP and cVAE. The superior performance of MetaDiffusion-Att can be attributed to its improved network architecture and the incorporation of attention mechanisms, which allow it to learn more effectively from small datasets. In contrast, MetaDiffusion's performance is somewhat limited by the relatively small training dataset.

Although we use the results after binarization as the outputs of the models, directly observing the outputs of the models can reflect the generation ability and stability of the model to a certain extent. From the figure, both MetaDiffusion-Att and MetaDiffusion can directly output clear structural images, while WGAN-GP has some blurring phenomena in its generated results due to training instability caused by adversarial processes. The cVAE also tends to generate blurry images, which is because VAE needs to balance the image reconstruction error and the KL divergence between the latent space and the standard distribution during training. This inherent problem limits the cVAE to generate stable and uniform patterns.

Figure 7 quantitatively compares the accuracy of the four models in generating meta-atoms according to transmission amplitude labels on the entire test set. Figure 7(a) shows the mean MAE on the entire test set. The average error of MetaDiffusion-Att on the entire test set is 0.02627, which is significantly lower than the average errors of MetaDiffusion (0.04946), WGAN-GP (0.06981) and cVAE (0.09575). MetaDiffusion-Att has a 47%, 62% and 73% accuracy improvement compared to MetaDiffusion, WGAN-GP and cVAE, suggesting its superior accuracy in inverse meta-atom design.

We also examine error distributions to avoid instances where a few high-performing samples might skew the overall performance assessment. Figure 7(b) illustrates these distributions, where green, blue, orange, and purple bars represent the error proportions for MetaDiffusion-Att, MetaDiffusion, WGAN-GP, and cVAE, respectively. The solid lines, obtained using kernel density estimation (KDE), visualize the overall distribution trends. As evident from Fig. 7(b), the error distributions of MetaDiffusion-Att and MetaDiffusion are more concentrated towards zero, indicating better overall performance. In contrast, WGAN-GP and cVAE exhibit more uniform distributions across the error range. To quantify these differences, we mark the 90th percentile performance for each model with a dashed line. Our proposed MetaDiffusion-Att model achieves a 90th percentile performance of 0.061, significantly outperforming the other models: MetaDiffusion (0.154), WGAN-GP (0.187), and cVAE (0.221). This translates to improvements of 61%, 67%, and 72% in the 90th percentile performance compared to MetaDiffusion, WGAN-GP, and cVAE, respectively.

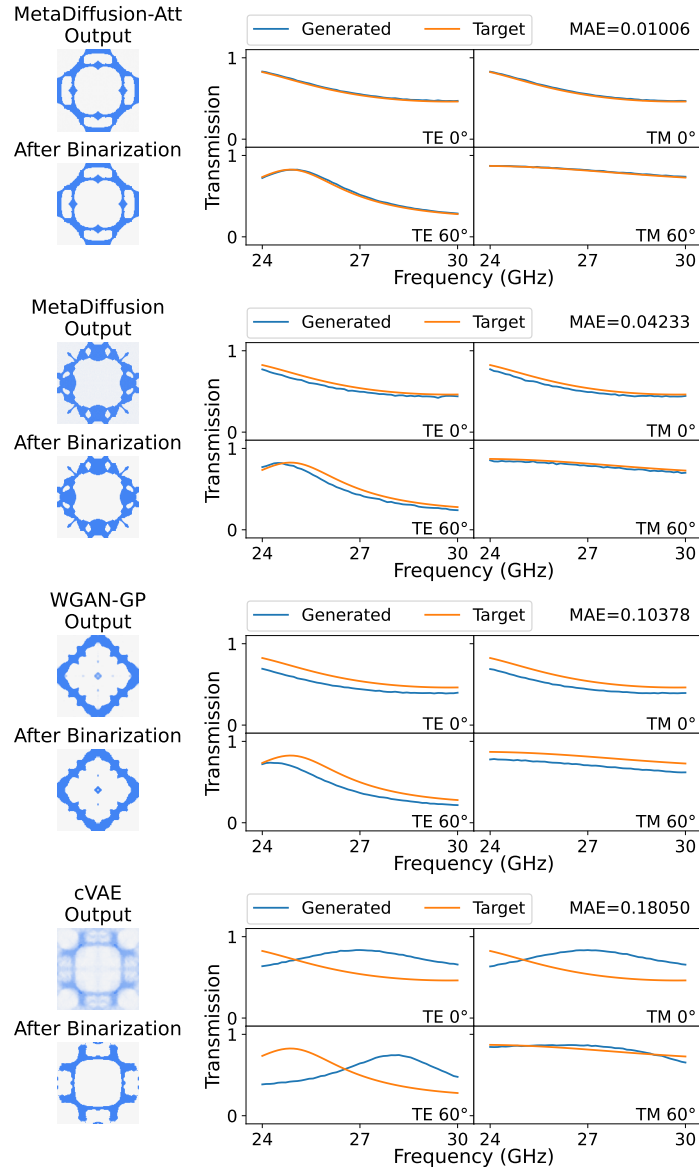


Fig. 5. Qualitative comparison example 1 of on-demand inverse design on a randomly selected sample from the test set. The y-axis represents transmission amplitude. In the structural representation, white indicates the tri-silver Low-E coating, while blue denotes the glass substrate exposed by etching. This color coding is used solely to enhance the visualization of the structure; it should be noted that in reality, both the tri-silver coating and the glass substrate are nearly optically transparent.

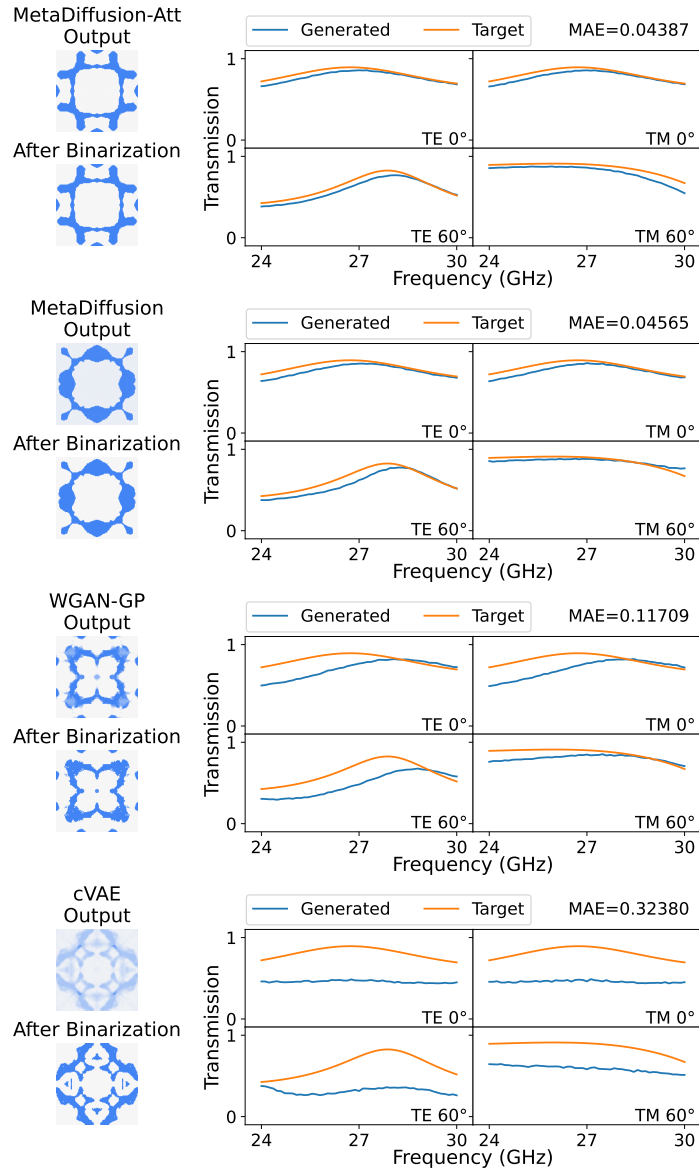


Fig. 6. Qualitative comparison example 2 of on-demand inverse design on a randomly selected sample from the test set. The y-axis represents transmission amplitude.

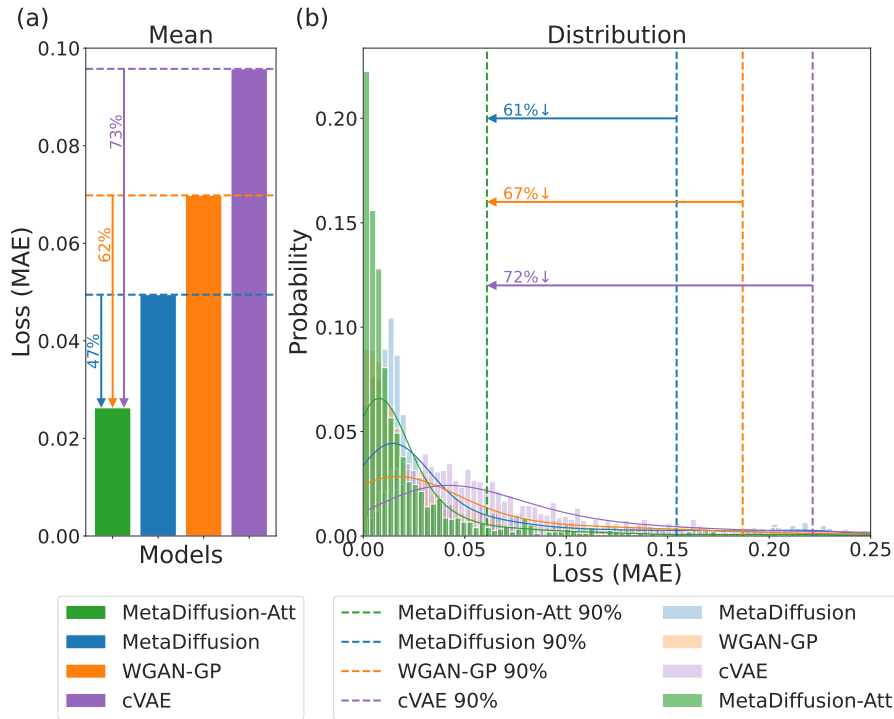


Fig. 7. Quantitative comparison of the four methods. (a) Mean MAE error on the test set. The improvement of MetaDiffusion-Att relative to original MetaDiffusion, WGAN-GP and cVAE is indicated by arrows in the figure. (b) Error distribution on the test set. The vertical dashed line indicates the position corresponding to the 90th percentile performance.

Having validated the effectiveness of our multi-objective optimization-based data collection method in improving distribution uniformity and the accuracy of our MetaDiffusion-Att model, we now demonstrate the capabilities of our integrated approach in the specific task of electromagnetic window design. Figure 8 illustrates that our method is capable of extrapolating from samples obtained by multi-objective optimization algorithms. This capability not only yields results that surpass those in the dataset but also enhances the diversity of the design space, effectively meeting a variety of design requirements. Figure 8(a) and 8(b) display examples where performance enhancements are achieved through extrapolation at normal incidence and at a high angle of 60 degrees incidence, respectively. By comparing these newly generated structures with the nearest samples in the design space, it is apparent that the network has, to some extent, learned the relationship between structure and electromagnetic response. This understanding enables precise adjustments to the microstructures, facilitating the necessary performance improvements. Figure 8(c) further illustrates examples of performance enhancements in both normal and high-angle incidences. Compared to the counterpart from the dataset, the new structures show significant improvement, exhibiting exceptional bandwidth across the entire range of 0 to 60 degrees incidence. These examples highlight our model's ability to not only interpolate between known designs but also extrapolate to create novel structures with performance beyond the original dataset limits. By generating structures with performance that exceeds the boundaries of the initial design space, our approach effectively expands the accessible design space for electromagnetic windows. This extrapolation capability has significant practical implications,

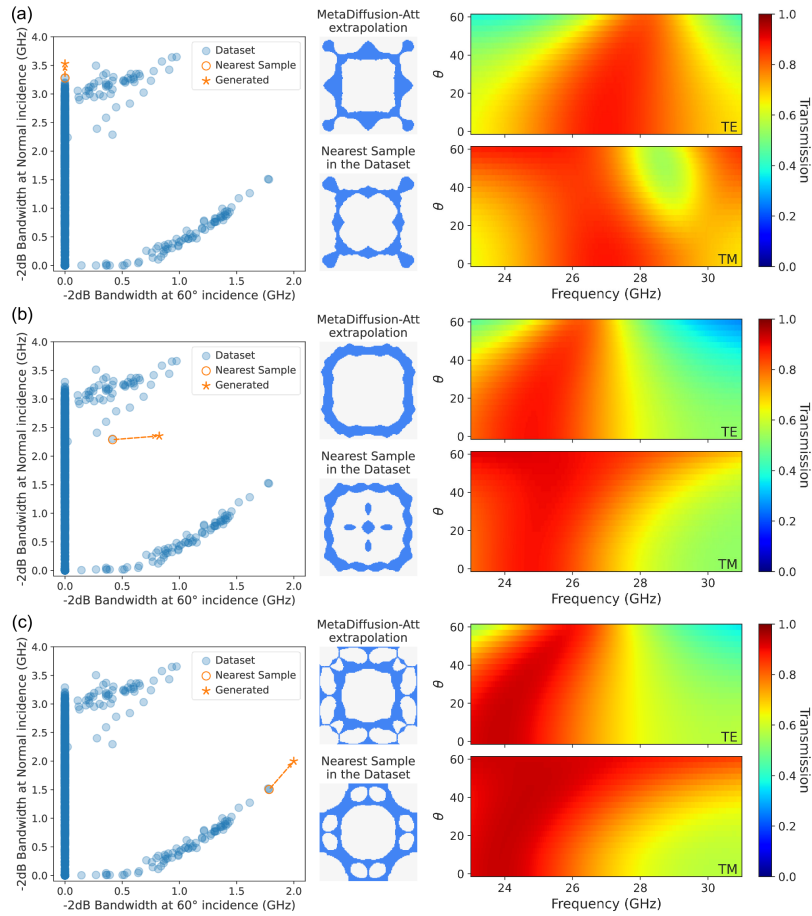


Fig. 8. Examples of extrapolation capability. From top to bottom, the panels illustrate examples of performance improvement at normal incidence, at high angle incidence, and simultaneously at both normal and high angles of incidence. Each newly generated structure is simulated using an electromagnetic solver to evaluate the transmission amplitude response across an incidence range from 0 degrees to 60 degrees.

potentially accelerating the development of high-performance electromagnetic windows for various applications, from telecommunications to energy-efficient building materials.

4. Conclusion

In conclusion, our study introduces the MetaDiffusion-Att framework, a novel inverse design approach that strategically combines multi-objective optimization with an enhanced diffusion model integrated with an attention mechanism. This framework addresses the significant challenge of sparse high-performance data availability in the design of complex electromagnetic devices, particularly Low-E glass with high degrees of freedom.

Our approach begins by utilizing a multi-objective optimization algorithm to effectively capture high-performance samples from limited data, a critical improvement over traditional dataset construction methods that often suffer from label imbalance. The optimized dataset subsequently trains the MetaDiffusion-Att model, which significantly outperforms established generative models like WGAN-GP and conditional VAE in terms of accuracy and quality, particularly in

scenarios with small datasets. Notably, the attention mechanism within the diffusion model architecture enhances feature learning, thereby improving generation precision and stability.

Through rigorous experimental validation, the MetaDiffusion-Att model not only demonstrates superior generation capabilities but also exhibits strong extrapolation capabilities. It successfully generates designs that exceed the performance parameters of existing samples within the training dataset, effectively expanding the design space and offering new possibilities in meta-atom generation.

The success of MetaDiffusion-Att in these complex design scenarios highlights its potential as a robust tool for exploring intricate design spaces and fulfilling diverse design requirements, especially in fields where high-performance samples are scarce. While our case study focuses on transmission magnitude, it's important to note that our approach is equally applicable to phase prediction or other electromagnetic properties. These could be incorporated as additional conditions for the network with minimal modifications to the overall architecture, further demonstrating the versatility of our method in addressing various aspects of electromagnetic design challenges.

Looking ahead, we aim to further enhance the efficiency of our model and explore its applicability to other challenging design tasks in nanophotonics and related fields. Future work could extend our approach to address emerging challenges in optical computing [45–47] and topological photonics [48]. These extensions could potentially contribute to the development of next-generation photonic devices that leverage both real-space and momentum-space engineering, thereby advancing the integration of electromagnetic functionalities with energy-efficient materials.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See [Supplement 1](#) for supporting content.

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